

Constructing Extra Stuff in a Diagram

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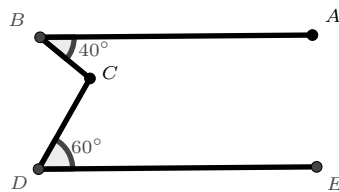
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"Sometimes beautiful things come into our lives out of nowhere. We can't always understand them, but we have to trust in them." — Lauren Kate

1 Prelude

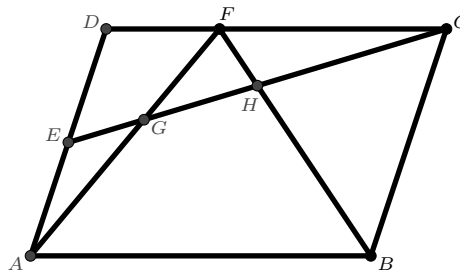
Perhaps the very first geometry problem that requires one to construct something extra is the following F.1 bookwork:

In the figure, $AB \parallel DE$, $\angle CBA = 40^\circ$ and $\angle EDC = 60^\circ$. Find the value of $\angle BCD$.



Then, when you go to local competitions you may probably face problems like

(HKMO 2016 Heat (Individual) Q14) As shown in the figure, $ABCD$ is a parallelogram. E is the midpoint of AD and F is a point on DC such that $DF : FC = 1 : 2$. FA and FB intersect EC at G and H respectively. Find the value of $\frac{\text{Area of } ABCD}{\text{Area of } \triangle FGH}$.



or perhaps

(IMC 2014 Individual Sec B Q2) In triangle ABC , $\angle ABC = \angle ACB = 30^\circ$. O is a point such that $OA = OC = 1$ and $OB = 2$. Moreover, $\angle CAB = \angle CAO + \angle OAB$. What is the length of BC ?

So you can see, constructing something extra is essential in olympiads. While the ones above in junior competitions could be easily ~~drawn accurately and measured~~ trained to add new points (say there are only a few standard tricks regarding area problems), those in high school olympiads are in general more difficult to think of. Yet the problems above sort of hints you the general philosophy behind constructing extra stuff. In this lesson, a few general principles would be introduced.

2 Better Interpretation of Condition/Statement

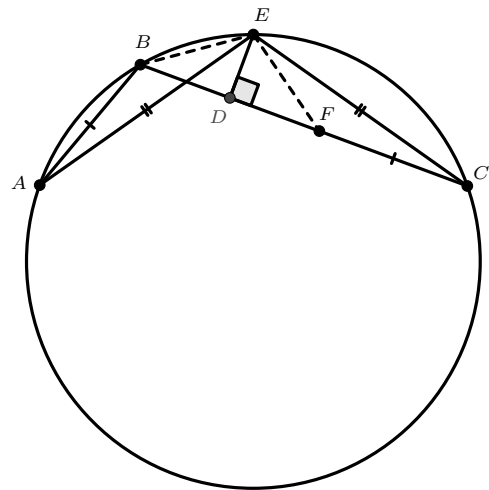
Sometimes the conditions or the problem statement of a geometry problem could be very weird, in the sense that it seems to be out of nowhere and rather difficult to be applied. Then you know it's probably some ingenious interpretation of the condition. One shall try to construct extra stuff in the diagram simply in order to make the condition useful.

Example 1. A, B, E, C lies on a circle in this order such that E is the midpoint of the arc $ABEC$. D is a point on chord BC such that $DE \perp BC$. Prove that

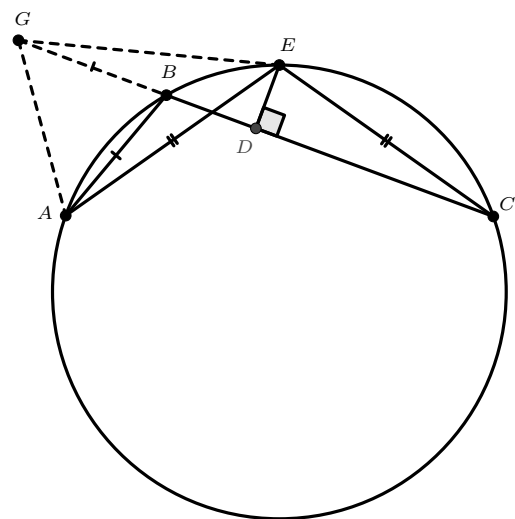
$$AB + BD = DC.$$

Solution. It's rather difficult to directly compare the lengths. Hence one shall try to find some ways to make the LHS and RHS of the equation *comparable*.

Method 1:



Method 2:



Problem 1 (IMO Shortlist 2007 G2). Denote by M midpoint of side BC in an isosceles triangle $\triangle ABC$ with $AC = AB$. Take a point X on a smaller arc MA of circumcircle of triangle $\triangle ABM$. Denote by T point inside of angle BMA such that $\angle TMX = 90$ and $TX = BX$.

Prove that $\angle MTB - \angle CTM$ does not depend on choice of X .

Problem 2 (IMO Shortlist 2003 G3). Let O be the circumcenter and H the orthocenter of an acute triangle ABC . Show that there exist points D , E , and F on sides BC , CA , and AB respectively such that

$$OD + DH = OE + EH = OF + FH$$

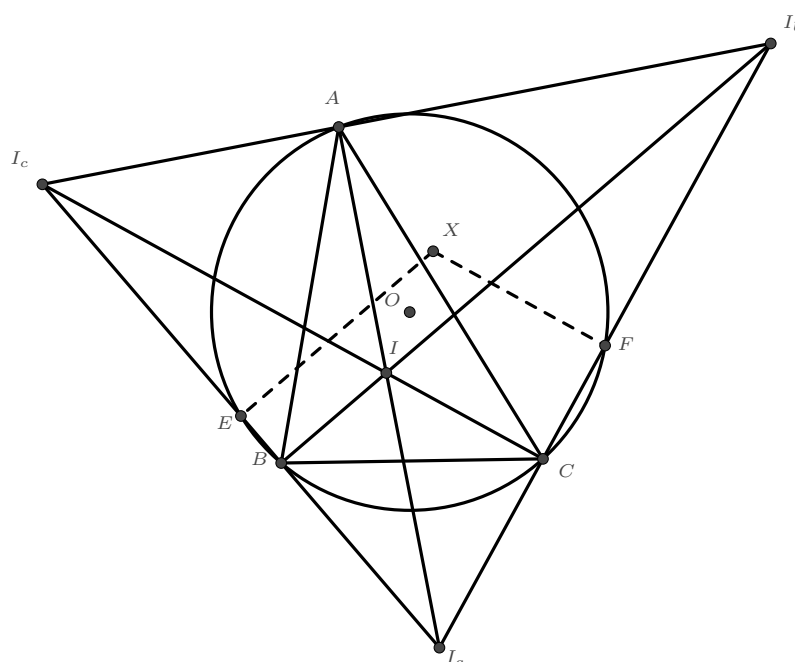
and the lines AD , BE , and CF are concurrent.

3 Complete the Figure! — Recognising some known configurations

How did one propose problems? It's often erasing some parts of a complete diagram. By "complete the figure" we mean recognising the diagram as a part of some sort of better known figures, so that we can make well use of some properties that are originally difficult to derive.

Example 2. In a triangle ABC , let I and O be its incentre and circumcentre. Denote by E the mid-point of arc ABC , and F the midpoint of arc ACB . Let the line passing through E perpendicular to BE and the line passing through F perpendicular to CF meet at X . Prove that I , O and X are collinear.

Solution. It's not likely that this problem could be solved by Menelaus' Theorem. It's not even likely to be solved by angle chasing since we know few properties about the line OI . To solve this problem, there must be some tricks.



Problem 3 (CHKMO 1998 Q1). In a concyclic quadrilateral $PQRS$, $\angle PSR = 90^\circ$. If H, K are the perpendicular feet from Q to sides PR, RS , prove that HK bisect segment SQ .

Problem 4 (CWMI 2010 Q2). AB is a diameter of a circle with center O . Let C and D be two different points on the circle on the same side of AB , and the lines tangent to the circle at points C and D meet at E . Segments AD and BC meet at F . Lines EF and AB meet at M . Prove that E, C, M and D are concyclic.

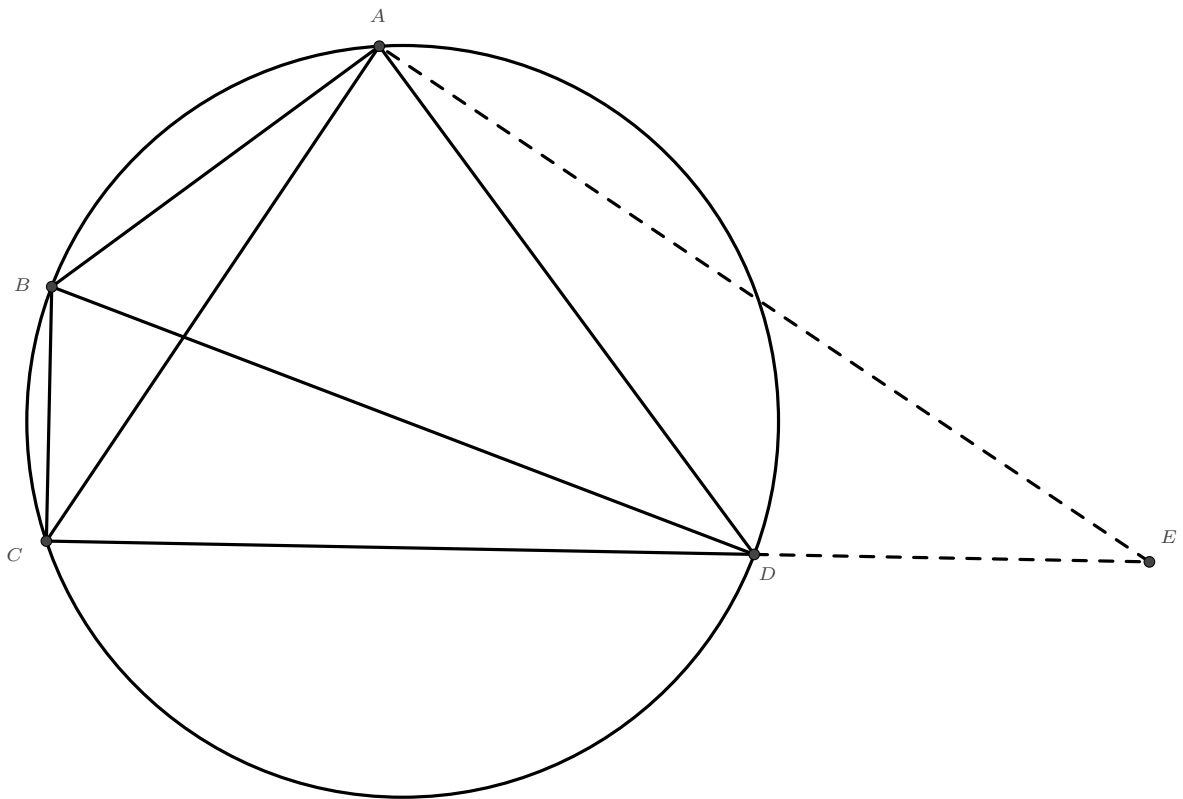
4 Transformations

The new points shall be constructed meaningfully so that they would turn out being useful. One of the most useful ones would be those out from transformations, since they give much more information than just how they are constructed. Say homothety, which give more than one pair of similar triangles upon one single new item constructed, are often noticed and employed. Hence, recognising the existence of some certain transformations is essential, since one can construct new points based on the transformation.

Example 3 (Ptolemy's Theorem). If A, B, C, D are concyclic in this order, prove that

$$AB \cdot CD + BC \cdot AD = AC \cdot BD.$$

Solution. If one does not wish to bash everything by brute force, one must construct some extra stuff to help you. Now the products is difficult to visualise in a figure, so we divide the equation by one of the quantities, say AB , hoping that CD , $BC \cdot \frac{AD}{AB}$ and $\frac{AC}{AB} \cdot BD$ are some legit lengths that actually appears in the figure for us to add them up.



Problem 5 (IMO Shortlist 2006 G2). Let $ABCD$ be a trapezoid with parallel sides $AB > CD$. Points K and L lie on the line segments AB and CD , respectively, so that $AK/KB = DL/LC$. Suppose that there are points P and Q on the line segment KL satisfying

$$\angle APB = \angle BCD \quad \text{and} \quad \angle CQD = \angle ABC.$$

Prove that the points P , Q , B and C are concyclic.

Problem 6 (IMO 2005 Q2). Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and BC not parallel with DA . Let two variable points E and F lie on the sides BC and DA , respectively and satisfy $BE = DF$. The lines AC and BD meet at P , the lines BD and EF meet at Q , the lines EF and AC meet at R .

Prove that the circumcircles of the triangles PQR , as E and F vary, have a common point other than P .

5 Coda

Those above are just some general advice. In fact, there are some standard tricks that one shall often check them out as sort of reflex action whenever some typical scenario appears, such as

- Midpoints: while it's in general difficult to deal with midpoints and angles together, one often
 - Construct another midpoint so that the midpoint theorem could be used; or
 - Reflect another point about this midpoint so that a parallelogram is formed
- A bunch of circles appearing in the figure, possibly intersecting with each other:
 - Construct their pairwise radical axis and mark some concurrent points; or
 - Construct the homothety centres, especially when there exist tangent circles
- Spiral similarity: it's more than just a pair of similar triangles because
 - One shall mark another pair of similar triangles, and draw them out; or
 - Mark the intersection of corresponding sides, which will induce some concyclicity; or
 - Mark the corresponding points for the spiral similarity and recognise it as another pair of stuff with spiral similarity, then repeat the process above
- Collinearity of some centres, which one's hopeless by Menelaus':
 - Recognising the centres as some other centres of some other triangles, and draw the corresponding properties of that centre wrt to our new triangle; or
 - Marking some possible homothety centres or inversion centres and try to transform one point to another in order to establish collinearity

etc. It's apparently never an exhaustive list. They all appear "naturally" upon the willingness for a better interpretation of the condition or simply a better big picture. When one learns some geometry theorems, it's important to recognise the configuration and recalls it whenever it looks alike in the future. After all, a technique will change from "out from nowhere" to "standard" only when one has faced sufficiently many problems using this technique!

Problem 7 (CGMO 2013 Q2). $ABCD$ is a trapezoid with $AB \parallel CD$. The sides DA , AB , BC are tangent to $\odot O_1$ and AB touches $\odot O_1$ at P . The sides BC , CD , DA are tangent to $\odot O_2$, and CD touches $\odot O_2$ at Q . Prove that the lines AC , BD , PQ meet at the same point.

Problem 8 (IMO 2010 Q2, proposed by Hong Kong). Given a triangle ABC , with I as its incenter and Γ as its circumcircle, AI intersects Γ again at D . Let E be a point on the arc BDC , and F a point on the segment BC , such that $\angle BAF = \angle CAE < \frac{1}{2}\angle BAC$. If G is the midpoint of IF , prove that the meeting point of the lines EI and DG lies on Γ .

Problem 9 (CMO 2017 Q2). In acute triangle ABC , let $\odot O$ be its circumcircle, $\odot I$ be its incircle. Tangents at B, C to $\odot O$ meet at L , $\odot I$ touches BC at D . AY is perpendicular to BC at Y , AO meets BC at X , and OI meets $\odot O$ at P, Q . Prove that P, Q, X, Y are concyclic if and only if A, D, L are collinear.